

Objects 0x3000 to 0x32A3: Elmo parameters objects

The 0x3000 to 0x32A3 objects can be used to address any of Elmo's parameters via SDO. The reference list from Elmo command to object index is indicated in the Gold Command Reference.

Note that the object is relevant only if the alias Elmo's command exists. Otherwise an appropriate Abort message is transmitted.

The maximum sub-index that can be accessed via objects 0x3XXX is 255. Some Elmo parameters have more sub-indexes which cannot be accessed via the objects.

Refer to [Chapter 4: Addressing Elmo Parameters via CANopen Objects](#) for more details.

- Object general description

Index	0x3000 – 0x32A3
Name	Elmo parameters objects
Object code	ARRAY
Data type	UNSIGNED32
Category	0

- Entry Description

Sub-index	0
Description	Highest sub-index supported
Entry category	Mandatory
Access	Read only
PDO mapping	No
Value range	None
Default value	None

Sub-index	1...255
Description	Local Error Reaction
Entry category	Optional
Access	R/W
PDO mapping	No
Value range	$0 \dots (2^{32}) - 1$
Default value	None

Note that a specific object and the sub index is subjected to the restrictions of the alias Elmo command. Please refer to the Command reference manual.

When access to object 0x30XX via CANopen SDO fails, the drive returns an SDO abort message that contains the SDO abort code. Detailed reason of the abort can be retrieved with uploading object 0x2081.4 or **EE[4]** via terminal.

Example 1

Try to set "MO=1" by sending **Object 0x3146** sub index 0 value 1 under condition of undervoltage. In this case the drive returns SDO abort message "General error", **EE[4]** and **0x2081.4** return error 233 "Undervoltage"

The following are the CANopen messages:

Client sends MO=1

601 SDO 22 46 31 00 01 00 00 00

Server responds with abort message "General error"

581 80 46 31 00 00 00 00 08

Client retrieves object 0x2081.4 (alias EE[4])

601 SDO 40 81 20 04 00 00 00 00

Server responds "Undervoltage"

581 43 81 20 04 E9 00 00 00

Example 2

Try to set UI[50]=200 by sending **Object 0x3120** sub index 50 value 200 (UI max sub index is 24). In this case the drive returns SDO abort message, **EE[4]** and **0x2081.4** return error 3 "Sub index does not exists"

The following are the CANopen messages:

Client sends UI[50]=200

601 SDO 22 20 31 32 C8 00 00 00

Server responds with abort message "Sub index does not exist"

581 80 20 31 32 11 00 09 06

Client retrieves object 0x2081.4 (alias EE[4])

601 SDO 40 81 20 04 00 00 00 00

Server responds "Bad index"

581 43 81 20 04 03 00 00 00